Table 1: Core AIOPs features in OpenSearch

Feature	Description	Plugin
Monitoring	Visualisations of logs/ metrics, Dashboards	OpenSearch Dashboards
Predictive insights	Trend detection via time series machine learning models	ML Commons
Anomaly detection	Real time machine learning models for anomaly detection	ML Commons
Log ingestion	Log parsing from various sources	Data Prepper
Alerting	Provides triggers on various anomalies-related events	Alerting plugin

The main advantages of this tool are:

- It is open source and free to use.
- Provides easy integration with ELK Stack (Elasticsearch, Logstash, Kibana, Beats) tools, and is fully compatible with Logstash and Beats.
- It has a built-in ML engine with the RCF (Random Cut Forest) algorithm.
- Its enhanced multi-variate anomaly detection capability helps detect data drift and unusual patterns automatically.

OpenSearch for anomaly detection

Large scale system monitoring can be done automatically and in real time with OpenSearch's integrated, AI-driven anomaly detection. Figure 1 describes the core components of the tool that can be used for anomaly detection.

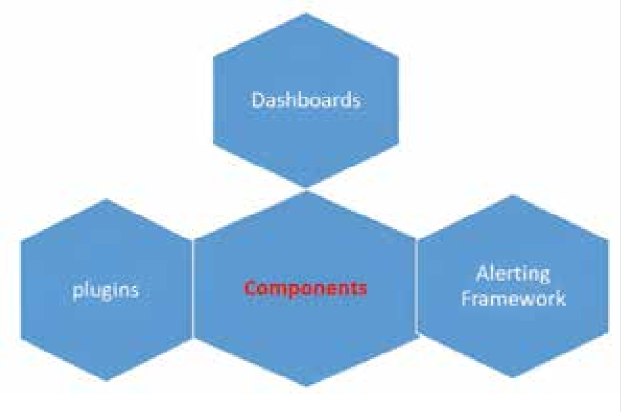

Figure 1: Core components of OpenSearch for anomaly detection

Plugins: A core plugin in OpenSearch is used to detect anomalies in time-series data using unsupervised machine learning. It uses a lightweight Random Cut Forest algorithm, which identifies data points that are far from the usual data points. This plugin has a multi-entity detection utility, which detects anomalies per host, service, etc, in real time. It uses REST API and UI-based management via dashboards, which visualise anomalies in an efficient way.

An index management plugin manages the data life cycle of indexes used for anomaly detection. This plugin defines policies for index rollover, retention, deletion, etc. It prevents storage bloat from continuous anomaly logging.

A performance analyser plugin is very helpful in anomaly root cause analysis. It analyses system level performance metrics, and correlates performance degradation with detected anomalies.

Dashboards: An online user interface tool called OpenSearch Dashboards lets you examine, view, and evaluate data kept in OpenSearch clusters. It is an OpenSearch Project fork of Kibana 7.10.2. Through user-friendly dashboards, charts, graphs, and plugins, it contributes significantly to security, monitoring, anomaly detection, and DevOps observability. The Anomaly Detection Dashboard offers a complete interface for controlling detectors and keeping an eye on any odd activity in data streams (Table 2).

Table 2: Various components of the Anomaly Detection Dashboard

Component	Description
Detector list	This list comprises 'View Status' and 'Type'.
Detector details view	This view comprises: • Real-time anomaly graphs • Historical anomaly heatmaps • Feature importance scores • Anomaly grade and confidence levels • Raw anomaly result data
Configuration panel	This has functionalities such as set features to monitor, define aggregation, choose detector interval, and delay and set filters for multivariate detection.
Linked visualisations	This helps to link anomaly results to external dashboards.

Alerting framework: The alerting plugin in OpenSearch is a powerful tool used to monitor your data and trigger actions when specific conditions are met. It makes it possible to receive real-time alerts about your metrics, logs, and anomaly detection findings. You may set up Monitors, Triggers, and Actions to alert you or automate corrective action when thresholds are exceeded or patterns are found.

The core concepts of the OpenSearch alerting framework are:

- *Monitor:* Specifies which data to monitor and how frequently. It runs a query or keeps an eye on an anomaly detector.
- *Trigger:* A condition (e.g., value exceeds threshold) within the monitor that specifies what to look for.